AMIRTESH RAGHURAM

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SUMMARY

Final-year B.Tech Biotechnology student (CGPA 9.01/10) with two first-author papers in *In Silico Pharmacology*. I build and validate computational drug discovery workflows that combine structure-based design, molecular dynamics, DFT, and machine learning, and I release them as open-source tools and deployed web applications.

EDUCATION

B.Tech Biotechnology | *Vellore Institute of Technology, Vellore* 2023 – 2027 | CGPA: 9.01/10

FIRST-AUTHOR PUBLICATIONS

- Raghuram A. "Computational characterization of Paraphamide B as a potential EGFR-preferential dual EGFR/ALK-binding candidate for non-small cell lung cancer." *In Silico Pharmacology*, 2026 (sole author). doi:10.1007/s40203-026-00777-3
  Filtered 36,454 NPAtlas natural products to 3,912, screened them with Smina, and characterized the lead with triplicate 200 ns GROMACS MD, MM/GBSA, ADMET profiling, and DFT (ORCA, B3LYP-D3BJ).
- Raghuram A. et al. "Computational Design and Validation of a Multi-Epitope Subunit Vaccine Against the Neglected Parasite *Blastocystis* sp." *In Silico Pharmacology*, 2026. doi:10.1007/s40203-026-00660-1

PROFESSIONAL EXPERIENCE

Summer Research Intern | *Strand Life Sciences, Bangalore* May – July 2025

- Automated Bash pipelines for NGS data processing, quality control, and clinical variant verification, cutting manual analysis time by ~80%.
- Ran the BLAST command-line suite across 50+ genomic samples to identify homologous regions in diagnostic workflows.

RESEARCH PROJECTS

AD-MTDL: Multi-Task D-MPNN and REINVENT4 RL Bridge for Alzheimer's Drug Design | GitHub

- Trained a multi-task D-MPNN (ChemProp v2) on 11,297 ChEMBL 36 compounds across four targets (AChE, BuChE, BACE1, MAO-B), reaching AUROC 0.853–0.986.
- Built an HTTP/Flask bridge that resolves the ChemProp v2 and REINVENT4 incompatibility, enabling 5 reinforcement-learning generation campaigns (215k SMILES).

KinaseForge: Generative AI for Kinase Inhibitor Discovery | kinaseforge.vit.ac.in

- Fine-tuned NovoMolGen-157M to generate 2.8M novel kinase-like SMILES, and trained XGBoost QSAR models for 25 kinases.
- Deployed a searchable multi-target query platform on institutional infrastructure.

AI-AMP Suite: Antimicrobial Peptide Classification and De Novo Generation | Web App | GitHub

- Built a multi-label XGBoost AMP classifier (ESM2 + biophysical features; 14,817 sequences from DBAASP, dbAMP, DRAMP) with a binary gatekeeper model (AUROC 0.947; 0.912 on external LMPred data).
- Pre-trained SmallPeptideGPT (4.77M parameters) on 3.37M PeptideAtlas sequences for de novo peptide generation (93% novelty).

Ferroptosis Predictor: Interpretable QSAR Framework

- Classified 1,624 ferroptosis modulators using 2,092 RDKit descriptors and Morgan fingerprints (AUROC 0.86 on a Butina clustered split to limit scaffold leakage).
- Used SHAP to map 75 key features to functional-group motifs, with 91.3% alignment on ChEMBL compounds.

Structure-Based Virtual Screening: TNBC, Glaucoma, and Diabetes

- Screened natural compounds against multiple triple-negative breast cancer targets with AutoDock Vina, triplicate GROMACS MD simulations, MM/GBSA free energy, and ADMET profiling (SwissADME, pkCSM).
- Extended the pipeline to carbonic anhydrase II (glaucoma) and alpha-amylase (diabetes), validating leads with MD simulations and ADMET filters.

OPEN-SOURCE TOOLS AND APPLICATIONS

- **Torchify** – PyTorch utility library with 4,100+ PyPI downloads. GitHub
- **QUEST** – CLI for ligand strain energies (GFN2-xTB) that integrates tautomer enumeration (RDKit), pocket detection (P2Rank), and docking (Vina). GitHub
- **Pose-Rescorer** – MM/GBSA rescoring with Rapid Perturbation Sampling, validated on EGFR, HIV-1 PR, BRD4, and HSP90. Docs
- **DynaMune** – ENM/NMA protein dynamics platform (ProDy): PRS, domain-hinge, and interface stability analysis. GitHub
- **Blood-Brain Barrier Permeability Predictor** – ChemProp v2 MPNN on B3DB (7,807 compounds) with an MCC-optimized threshold (0.58). Live App
- **Automated-Virtual-Screening** – Parallel docking pipeline (Vina, Smina, QVina). GitHub

TECHNICAL SKILLS

Computational Drug Discovery: Structure-Based Drug Design, Molecular Docking (AutoDock Vina, Smina, QVina), GROMACS MD (CHARMM36, CGenFF), MM/GBSA and MM/PBSA (gmx_MMPBSA), DFT (ORCA), xTB and CREST, ADMET Profiling (SwissADME, pkCSM, ProTox 3.0), In Silico Vaccine Design

Cheminformatics and ML: RDKit, QSAR Modeling, Generative Molecular Design, ChemProp (D-MPNN), XGBoost, SHAP, PyTorch, Scikit-learn

Programming: Python (NumPy, Pandas, BioPython), R, Bash, Git, Flask, Streamlit

Structural Bioinformatics and Genomics: ProDy, PyMOL, UCSF Chimera, Discovery Studio, AlphaFold2/3, SWISS-MODEL; NGS pipelines and BLAST

Working Knowledge: RNA-Seq, scRNA-Seq, ChIP-Seq, Variant Calling (GATK), GWAS (PLINK), DNA Methylation, Somatic Mutation Profiling, TensorFlow, Bio3D, I-TASSER